

29th RoboRacer Autonomous Racing Competition

Technical Documentation

Vehicle Hardware and Software Stack

Team AMSLab

Keimyung University, Republic of Korea

IFAC 2026, BEXCO Busan · August 24-27, 2026

July 2026

1. Team Information

Item	Detail
Team name	AMSLab
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2. Vehicle Hardware

2.1 Platform Overview

The vehicle is a 1/10-scale autonomous racing platform built according to the official RoboRacer (F1TENTH) build guideline, based on a Traxxas 4WD rally chassis. All computation is performed on board by a single main computer; wireless connectivity is used exclusively for visualization and debugging and no data is transmitted to the vehicle during autonomous runs.

Component	Model	Notes
Chassis	Traxxas Ford Fiesta ST Rally BL-2s (Model 74154-4), 1/10 scale, 4WD	Length 535 mm, track width 281 mm, wheelbase 324 mm
Drive motor	Hobbywing XeRun V10 G4R 17.5T (3100KV)	Sensored brushless, single motor
ESC	Trampa VESC 6 MKVI HP (Higher Power edition, 12 FETs)	FOC drive, wheel odometry (ERPM), servo PWM output, built-in IMU
Steering servo	HPS CB700 S.Bus High Performance Brushless Surface Servo	Steering actuation
Servo power	Hobbywing UBEC 10A (2-6S)	Regulated supply for the steering servo
Battery	DXF 3S LiPo, 11.1 V, 7200 mAh, 100C (single pack)	Within the 4S regulation limit
Power distribution	F1TENTH Power Board v2024.1 (Ambimat Electronics)	Buck-boost rails: 19 V (main computer), 12 V (LiDAR), 5 V (accessories)
Main computer	GEEKOM Mini GT13 Max (Intel Core Ultra 9 185H, x86_64)	Single onboard compute unit
LiDAR	Hokuyo UST-20LX 2D scanning laser rangefinder	270 deg FOV, 0.25 deg resolution, 20 m guaranteed range, single unit
Camera	Not used	-
Wireless	Wi-Fi (onboard adapter + external antenna)	Visualization and debugging only

2.2 Power Architecture

A single 3S LiPo pack supplies the whole vehicle. The VESC 6 MKVI HP is connected directly to the battery and drives the sensed brushless motor using field-oriented control. The F1TENTH Power Board v2024.1 distributes battery power to the onboard electronics: a 19 V rail powers the main computer and a dedicated 12 V header powers the Hokuyo LiDAR. The steering servo is powered through a Hobbywing UBEC 10A to provide a stable, isolated supply independent of the drive electronics.

2.3 Regulation Compliance Summary

Regulation item	Requirement	Our vehicle
Chassis dimensions	Width 238-341 mm, Length 454-654 mm	281 mm x 535 mm (compliant)
Drivetrain	Single motor	One XeRun V10 G4R 17.5T
Main computer	Single compute unit	One GEEKOM Mini GT13 Max
LiDAR	Single unit	One Hokuyo UST-20LX
Camera	No onboard detection / VIO output	No camera used
Battery	4S LiPo maximum	3S LiPo x 1
Computation	All computation onboard	Fully onboard, no external offloading
Emergency stop	Toggle (on/off) e-stop demonstration	Toggle-type autonomous/manual mux with e-stop
Head-to-head accessories	Detection box and foam bumper when 2+ cars on track	Detection box ($\geq 12 \times 12$ cm over the 10-30 cm height band) and foam bumper prepared

3. Software Stack

The autonomy software is based on the open-source UNICORN racing stack (HMCL-UNIST) running on ROS 2 Jazzy, adapted and tuned by the team for our hardware configuration. The stack follows a modular pipeline: perception, tracking, prediction, planning, behavior state machine, and control. Software-in-the-loop verification is performed in the included f1tenth_gym-based simulator prior to on-vehicle testing.

3.1 Mapping

- An occupancy grid map of the track is built offline with Google Cartographer during the mapping session.
- The track centerline and boundaries are extracted from the map image and used as the reference for global trajectory optimization and the Frenet coordinate frame.

3.2 Localization and State Estimation

- Online localization uses a Particle Filter (Monte Carlo Localization) with a ray-casting measurement model accelerated by range_libc, operating on the prebuilt occupancy grid map.

- The motion model is driven by wheel odometry derived from VESC ERPM feedback (with adaptive gain calibration) fused with IMU measurements.
- Cartographer pure localization is retained as an alternative localization mode.

3.3 Global Planning

- A minimum-curvature global raceline is optimized offline from the extracted track boundaries.
- A velocity planner assigns a curvature- and friction-aware speed profile along the raceline, tuned for the low-friction urethane-coated concrete surface of the venue.

3.4 Perception: Detection

- Incoming LiDAR scans are filtered against the static occupancy grid (grid/map filter) so that only points not belonging to the track remain.
- The residual points are clustered and passed through feature extraction to produce obstacle candidates with position, size, and orientation. The detection node is implemented in C++ for real-time performance.

3.5 Perception: Tracking and Prediction

- Detected objects are associated over time and tracked in the Frenet frame with an Extended Kalman Filter, distinguishing static obstacles from a dynamic opponent.
- The opponent's future trajectory is predicted with a Gaussian Process-based trajectory predictor to support safe overtaking decisions.

3.6 Local Planning and Avoidance

- Static obstacles are avoided with a spline-based local planner (spliner) that generates a smooth evasive trajectory around the obstacle and rejoins the global raceline.
- Dynamic overtaking uses a lane-change planner that selects a collision-free lane based on the tracked and predicted opponent state.
- A recovery planner returns the vehicle to the raceline after off-nominal situations.

3.7 Control

- Lateral control: Pure Pursuit with adaptive (L1-based) lookahead scaling with speed, with curvature- and lateral-error-dependent speed scaling for stable high-speed cornering.
- Longitudinal control: tracking of the planned velocity profile, with a PID-based trailing controller that maintains a safe gap when following the opponent.
- A Follow-the-Gap reactive controller is available as a fallback driving mode.

3.8 Behavior State Machine

- A rule-based state machine supervises the driving behavior, switching among global tracking, trailing, overtaking, and recovery states based on the perception and tracking outputs.

3.9 Safety

- The autonomous/manual multiplexer uses a toggle (on/off) switch scheme as required by the software inspection; the toggle e-stop immediately overrides all autonomous commands.
- The vehicle operates independently of Wi-Fi status; wireless connectivity is used only for visualization and debugging, and no data is sent to the vehicle during runs.

4. Head-to-Head Accessories

- Detection box: occupies at least 12 x 12 cm on every horizontal plane between 10 cm and 30 cm above the ground, so that the opponent's LiDAR can reliably detect the vehicle.
- Foam bumper: a soft foam front bumper is mounted to minimize damage in the event of contact.
- Both items are mounted whenever two or more vehicles are on the track, per regulation.

5. References

- Official competition rules: <https://2026ifac-roboracer.com/en/rules.html>
- RoboRacer build guideline: <https://roboracer.ai/build>
- UNICORN racing stack (base software): <https://github.com/HMCL-UNIST/unicorn-racing-stack>
- 2026 IFAC Busan tutorial: <https://hmcl-unist.github.io/unicorn-racing/wiki/2026-tutorial/>